Does Strengthening Rent Control Reduce Housing Quality? Evidence from New York City

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Abstract

This paper studies how New York's 2019 Housing Stability and Tenant Protection Act (HSTPA) affected housing quality and investment. HSTPA eliminated vacancy decontrol and sharply limited rent increases for capital improvements, weakening landlords' incentives to invest in rent-stabilized buildings. Using building-level panel data on housing code violations, tenant complaints, and building permits in a difference-in-differences design, I find that HSTPA reduced housing quality and investment: immediately hazardous violations increased by 36%, or approximately 23,000 additional violations per year, tenant complaints increased by 13%, and building permit activity declined by 25%. These effects are sharply unequal: immediately hazardous violations increased around three to four times more in lower-income neighborhoods and neighborhoods with higher non-white population shares. These results provide new evidence on the maintenance, investment, and distributional effects of modern rent regulation.

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1 Introduction

Rising housing costs have made affordability a central policy issue in high-demand housing markets (Galster and Lee, 2021; Baum-Snow and Duranton, 2025). Policymakers in many cities and states have responded by enacting or strengthening rent regulation (Ahern and Giacoletti, 2026).[1] A long-standing theoretical concern with these policies is that they reduce landlords' incentive to maintain housing: if future rents cannot reflect investments in quality, upkeep declines and the stock deteriorates (Friedman and Stigler, 1946). The empirical literature has established quality as a core margin of landlord response (Kholodilin, 2024), but credibly identifying the causal effect of rent regulation on quality is difficult. As cities continue to expand rent regulation, understanding the size, persistence, and incidence of any quality response is a first-order policy question.

This paper provides causal evidence on that question by studying the effects of New York's 2019 Housing Stability and Tenant Protection Act (HSTPA) on housing quality and investment in New York City. Before the law, landlords could raise rents after vacancies, recover capital-improvement costs through permanent rent increases, and eventually remove high-rent units from stabilization. HSTPA eliminated vacancy decontrol, sharply curtailed vacancy increases, and made capital-improvement recovery smaller and temporary, with the goal of limiting rent increases for tenants in rent-stabilized apartments. The reform's scale, its sharp restructuring of investment incentives at a single point in time, and the availability of detailed administrative data on New York City apartment buildings make this a particularly well-suited setting for studying landlord responses on the quality margin.

I assemble a rich building-level panel of New York City administrative records from 2016 through 2025. The data combine building characteristics with rent-stabilization status, housing code violations, tenant complaints, inspection-to-certification intervals, building alteration permits, neighborhood characteristics, information on landlords, and mortgage records. These outcomes capture complementary dimensions of housing quality: inspector-verified hazards, tenant-reported problems, administrative repair responsiveness, and capital investment. This breadth allows me to study not just whether violations rise, but whether the pattern across complaints, repair timing, and permits points to reduced maintenance and investment, and to identify which neighborhoods bear the largest effects.

Empirically, I compare the evolution of outcomes in rent-stabilized and non-stabilized buildings before and after HSTPA using a difference-in-differences design. The building-

[1] Recent examples include Oregon's SB 608, enacted in 2019, which limited annual rent increases to 7 percent plus CPI; California's 2019 Tenant Protection Act, which capped annual increases at 5 percent plus local CPI or 10 percent, whichever is lower; St. Paul's 2021 rent stabilization ordinance; and Montgomery County, Maryland's 2023 rent stabilization law.

level panel allows me to include building fixed effects, which absorb time-invariant building characteristics, and flexible time controls for neighborhood, building age, size, type, and baseline quality. Identification therefore comes from within-period comparisons between stabilized and non-stabilized buildings, after allowing outcomes to evolve differently across these neighborhood and property-characteristic groups.

I find that HSTPA worsened measured housing quality and reduced investment. Immediately hazardous Class C violations increased by 1.2 violations per 100 units per 6-month period, a 36% increase relative to the pre-HSTPA mean among stabilized buildings. This implies approximately 23,000 additional immediately hazardous violations per year across the city's stabilized stock. Tenant complaints increased by 13%, inspection-to-certification intervals increased by 34%, and building permit activity declined by 25%. The effects are concentrated in the most serious violation category: Class C violations rise substantially, while Class A violations rise little. Together, the rise in serious verified violations, the increase in tenant complaints, the longer certification intervals, and the decline in permit activity point to real deterioration rather than only increased reporting.

The effects are also sharply unequal. Class C violations increased by 1.72 per 100 units in low-income neighborhoods, compared with 0.59 in high-income neighborhoods. In neighborhoods in the top tercile of non-white population share, violations increased by 2.28 per 100 units, compared with 0.64 in neighborhoods in the bottom tercile. These differences suggest that the quality costs of stronger rent regulation are concentrated in lower-income and predominantly non-white neighborhoods.

Financial constraints amplify the response but do not fully explain it. Buildings with pre-HSTPA mortgages show substantially larger quality declines, consistent with the findings of Seltzer (2024), who shows that financing constraints reduce maintenance in rent-stabilized apartments. However, even buildings without a recorded pre-HSTPA mortgage experience significant deterioration. This pattern suggests that HSTPA reduced maintenance not only by tightening cash flow for leveraged owners, but also by lowering the return to preserving quality.

This paper contributes to the literature on rent regulation and landlord behavior. The modern empirical literature shows that rent control affects rental supply, tenant mobility, allocation, and the distribution of housing wealth. Diamond et al. (2019) find that San Francisco's 1994 rent-control expansion reduced rental supply through conversions and redevelopment. Autor et al. (2014) show that the end of rent control in Cambridge increased property values and investment, with spillovers to nearby units. Ahern and Giacoletti (2026) show that St. Paul's 2021 rent-control ordinance reduced property values and generated uneven wealth effects across the income distribution of renters. Rent control also generates tenant-unit misallocation and price spillovers to the unregulated segment (Glaeser and Luttmer, 2003; Mense et al., 2023).

Prior evidence on housing quality is substantial, but it faces recurring measurement and identification constraints. Gyourko and Linneman (1990) show that New York City's old rent controls are associated with lower rental housing quality, and Moon and Stotsky (1993) use a structural model to show that greater rent suppression in New York City was associated with faster deterioration. Sims (2007) finds that decontrol in Massachusetts reduced maintenance problems, though primarily for minor deficiencies. Breidenbach et al. (2022) find that Germany's rent brake temporarily reduced asking rents, with that effect fading after about a year, and lowered the probability that treated rental listings were advertised in good condition. The closest published paper is Seltzer (2024), who studies a 2011 Rent Act change that reduced cost recovery for apartment improvements in larger rent-stabilized buildings. Using New York City housing-code violations and a matched difference-in-differences design, he finds that violations increased after the reform, especially among buildings whose owners were financially constrained. This paper builds on that work by studying a broader reform of an existing rent-stabilization system, using richer building-level measures of tenant-relevant quality, and documenting unequal quality costs across neighborhoods.

The existing evidence on HSTPA is mainly descriptive. NYU Furman Center (2021) examine short-run trends in complaints, violations, and alteration filings after HSTPA and find little immediate change in complaints or violations, though alteration filings declined after an initial spike. More recently, NYU Furman Center (2026) descriptively document deteriorating operating margins and larger post-2021 increases in code violations among legacy buildings with at least 90 percent rent-stabilized units. A survey of owners and managers commissioned by industry groups reports that owners viewed HSTPA as making apartment renovations harder to finance, and documents sharp declines in applications for rent increases tied to building-wide and apartment-level improvements after the law (HR&A Advisors, 2024). These descriptive reports and surveys show that the financial and physical conditions of stabilized buildings are an important policy concern, but they do not isolate the causal effects of HSTPA. This paper fills that gap by estimating the causal effect of HSTPA on the quality and investment decisions of buildings that remained in the stabilized stock.

The rest of the paper proceeds as follows. Section 2 describes HSTPA and the investment incentives it changed. Section 3 describes the building-level panel. Section 4 presents the empirical strategy. Section 5 reports the main results, heterogeneity by neighborhood and financial constraints, and robustness checks. Section 6 concludes.

2 Policy Background

2.1 Rent Stabilization in New York City

New York City's rent stabilization system covers approximately one million apartments, primarily in buildings with six or more units, built before 1974.[2] Unlike strict rent control, rent stabilization allows annual rent increases set by the Rent Guidelines Board, typically 1–3% per year for lease renewals. The program is a clear example of what Arnott (1995) calls "second-generation" rent control: rent growth is capped, but landlords can still recover operating cost increases.

Before HSTPA, landlords could raise legal rents through several additional channels. Upon tenant turnover, owners could generally raise rents through a statutory vacancy lease. Owners could also recover the cost of approved building-wide improvements, such as boilers, roofs, elevators, and windows, through Major Capital Improvement (MCI) rent increases. The cost of apartment-level renovations could be recovered through Individual Apartment Improvement (IAI) increases, which before HSTPA were generally permanent and uncapped, with formulas based on building size. Finally, high-rent and high-income deregulation allowed units to leave stabilization when legal rents and, in some cases, tenant incomes exceeded statutory thresholds.

These provisions gave landlords multiple ways to raise rents and, through vacancy decontrol, to eventually exit the system entirely. The result was that, despite the existence of rent stabilization, landlords maintained incentives to invest in building quality to increase rents and attract higher-income tenants.

2.2 The Housing Stability and Tenant Protection Act (2019)

On June 14, 2019, New York enacted the Housing Stability and Tenant Protection Act. The law was designed to strengthen tenant security, limit large rent increases after turnover or improvements, and reduce pathways through which regulated units left the stabilized stock. For landlords, the law reduced the value of the rent-increase and deregulation channels described above.[3]

HSTPA first weakened the link between tenant turnover and future rent growth. Before the law, a vacancy could raise the legal rent by up to 20% for a two-year lease, with additional longevity increases in some cases. HSTPA eliminated these statutory vacancy and longevity

[2]Exact counts vary depending on data source and year. The NYC Rent Guidelines Board estimated 966,000 stabilized units in 2018; see its rent-stabilized building lists.

[3]For an overview of HSTPA's purpose and early debates, see NYU Furman Center (2021), *Housing Stability and Tenant Protection Act: An Initial Analysis of Short-Term Trends*.

increases and barred the Rent Guidelines Board from setting separate vacancy adjustments. It also repealed high-rent and high-income deregulation. Before HSTPA, a unit could leave stabilization if its legal rent reached the deregulation threshold, which was $2,774 in New York City in 2019, either when the unit became vacant or, for occupied units, when tenant income exceeded $200,000 for two consecutive years. After HSTPA, rent increases no longer created the same path from investment to deregulation.[4]

HSTPA also sharply reduced landlords' ability to recover capital improvement costs, shrinking the rent increases owners could charge and, for both improvement channels, making those increases temporary rather than permanent. For building-wide Major Capital Improvements, the law capped the annual increase at 2% of existing rent and required it to be removed after 30 years. For Individual Apartment Improvements, as enacted in 2019, it capped eligible costs at $15,000 over a 15-year period, limited owners to no more than three increases in that window, and cut the monthly rent-increase formula to 1/168 or 1/180 of eligible costs.[5]

Taken together, these provisions strengthened tenant protections while weakening the link between building investment and future rental income. They reduced the expected payoff to capital improvements and the option value of improving units toward deregulation. This paper studies the effects of these changes on investment and routine maintenance in rent-stabilized buildings.

2.3 Investment Incentives and Empirical Predictions

By reducing the expected return to maintaining and improving rent-stabilized buildings, these provisions generate two empirical predictions. First, if the reduced-return channel dominates legal maintenance obligations and enforcement, HSTPA should reduce capital investment in stabilized buildings, especially discretionary renovations whose payoff depends on future rent growth or deregulation. I measure this margin using building permits, which capture renovation and alteration work requiring Department of Buildings approval. Second, HSTPA may worsen routine maintenance. When future rent increases are less responsive to building quality, owners have weaker incentives to prevent deterioration or repair problems quickly. I measure this margin using serious housing code violations, tenant complaints, and the time

[4]For an HCR overview of the 2019 HSTPA changes, see NYS Homes and Community Renewal, *Strengthening New York State Rent Regulations: The Housing Stability and Tenant Protection Act of 2019.* For historical deregulation rent and income thresholds, see HCR Fact Sheet #36.

[5]For MCI rules, see HCR Fact Sheet #24. The HCR overview cited above summarizes the 2019 IAI changes. Later legislation modified the IAI rules effective October 17, 2024, raising the cap to $30,000 generally and up to $50,000 in specified cases, and making new IAI increases permanent; see HCR's summary of FY24 housing-law changes. I describe HSTPA here as the 2019 policy shock.

from inspection to certification.

3 Data

3.1 Data Sources and Variables

I assemble administrative data from multiple New York City agencies to measure housing quality, investment, and rent-stabilization status at the building level. I link datasets using the Borough-Block-Lot (BBL) identifier, a 10-digit code that uniquely identifies every tax lot in New York City.

Building characteristics. The Primary Land Use Tax Lot Output (PLUTO) database, published by the Department of City Planning and available through NYC Open Data, provides the universe of tax lots and their characteristics.[6] I use PLUTO to obtain total residential units, year built, building class, census tract, and lot area. PLUTO defines the sample frame: I begin with all lots and apply the restrictions described below.

Housing quality outcomes. Data on violations and complaints come from two New York City Department of Housing Preservation and Development (HPD) datasets, Housing Maintenance Code Violations and Housing Maintenance Code Complaints and Problems, accessed through NYC Open Data. HPD enforces the New York City Housing Maintenance Code.

When HPD verifies a violation, it assigns one of three standard hazard classes: Class A (non-hazardous), Class B (hazardous), or Class C (immediately hazardous).[7] Class A covers non-hazardous violations such as missing notices, signage, recordkeeping problems, or lower-severity maintenance conditions. Class B covers hazardous conditions affecting health, safety, or habitability, while Class C is reserved for immediately hazardous conditions, including heat and hot-water failures, lead-based paint hazards, missing window guards, mold, and pest infestations. I assign violations to the month of HPD inspection. I focus on Class B and Class C violations because they capture serious living-standard failures, and use Class A as a lower-severity comparison group.

I also use the time from inspection to certification as a proxy for landlord responsiveness. I compute this interval as the number of days from HPD inspection to the certification date recorded in the violations data. I set negative intervals to missing and compute the median

[6] NYC Open Data is the city's public data portal, available at `https://opendata.cityofnewyork.us/`.

[7] HPD correction deadlines vary by class and condition: Class A violations generally must be corrected within 90 days, Class B violations within 30 days, most Class C violations within 24 hours, and selected Class C categories within 14 or 21 days; heat and hot water violations require immediate correction. See NYC HPD, *Penalties and Fees*, `https://www.nyc.gov/site/hpd/services-and-information/penalties-and-fees.page`.

within each building-month, so the measure is defined only for building-months with at least one violation with a nonmissing, nonnegative certification interval.

Finally, I use complaint counts to measure tenant-reported housing problems. Complaints are filed through 311, online, or HPD Code Enforcement Borough Offices and describe conditions such as lack of heat or hot water, water leaks, or pest infestations. They capture reported conditions regardless of whether HPD ultimately verifies a code violation, and serve as a complementary measure to actual housing violations.

Investment outcomes. Building permit data come from the Department of Buildings (DOB), also accessed through NYC Open Data. NYC law requires permits for significant construction work: alterations affecting building use or occupancy, structural modifications, and major plumbing or electrical work. I focus on alteration permits (types A1, A2, A3), which capture renovations requiring regulatory approval. I combine data from both the DOB's traditional filing system and DOB NOW, the online portal launched in 2016 that handles the majority of filings.[8]

Routine maintenance (painting, plaster repair, replacing fixtures, minor repairs) does not require permits. This distinction is central to interpretation: permit activity captures major capital investment, while violations and complaints capture routine maintenance. I normalize permit counts to per-100-units, consistent with the violation and complaint outcomes.

For descriptive analysis of the MCI cost-recovery mechanism, I use Major Capital Improvement application data from the NYCDB project.[9] MCIs are building-wide improvements (boilers, roofs, elevators) for which landlords of stabilized buildings can seek rent increases.

Treatment variables. The baseline estimates use a binary, building-level treatment indicator. I define treated buildings as sample buildings in the 2017 Division of Housing and Community Renewal (DHCR) registry, which lists buildings with at least one rent-stabilized unit; control buildings are sample buildings not listed in that registry.[10] I extract BBLs from these PDFs and match to the PLUTO database. Using the 2017 registry ensures that treatment classification is determined before HSTPA's passage in June 2019, avoiding any concern that post-treatment changes in stabilization status could bias the estimates.

For robustness, I also collect data on the share of units that were rent-stabilized in each building as of 2018, which I use as a continuous treatment measure. These data are

[8] Using only the traditional system would show a spurious decline after 2016 as filings shifted to the online portal.

[9] The MCI application records were originally obtained from HCR/DHCR through a FOIL request by Winnie Shen and distributed through NYCDB; see `https://github.com/nycdb/nycdb`.

[10] The registry is published by the NYC Rent Guidelines Board as PDF lists of stabilized buildings (BBLs) by borough. Available at `https://rentguidelinesboard.cityofnewyork.us/resources/rent-stabilized-building-lists/`. I use the 2017 registry, archived at `https://github.com/austensen/dhcr-rgb-buildings`.

downloaded from the NYCDB project, and were originally compiled from NYC Department of Finance (DOF) tax bills.[11] These shares also confirm that the binary-treated group is meaningfully stabilized: although the 2017 registry includes any building with at least one rent-stabilized unit, the median stabilized-unit share among treated buildings is 95%.

Building-level leverage. To test whether effects operate through a debt channel, I use mortgage recordings from the Automated City Register Information System (ACRIS), NYC's public database of property transactions and mortgage documents.[12]

For each building, I identify the most recent mortgage origination recorded before HSTPA and divide the mortgage amount by the building's 2018 assessed value from PLUTO. This mortgage-to-assessed-value ratio measures relative leverage across buildings.

The measure is a proxy rather than a true current loan-to-value ratio: I observe origination amounts, not outstanding balances, and assessed values likely differ from market values. The median mortgage-to-assessed ratio in my sample is 1.85.[13] I split buildings with pre-HSTPA mortgages into terciles by this ratio; buildings without a recorded pre-HSTPA mortgage form the reference group.

Landlord identification. To test for portfolio spillovers across a landlord's stabilized and non-stabilized buildings, I link buildings to individual owners using HPD Multiple Dwelling Registrations data.[14] NYC law requires owners of buildings with three or more residential units to register annually with HPD, providing contact information including the name of the owner or head officer. I extract standardized owner names and link buildings that share common ownership. The match identifies 52,579 buildings (98% of the sample)[15] and 24,933 unique owners; 1,880 owners (7.5%) have "mixed portfolios" containing both stabilized and non-stabilized buildings. Appendix C provides details on the name standardization and portfolio classification.

Neighborhood characteristics. For heterogeneity analysis, I use tract-level median household income and non-white population share from the 2015–2019 American Community Survey 5-year estimates, accessed via the Census API.

[11] Data originally compiled by John Krauss from DOF tax bills; see `https://github.com/nycdb/nycdb`.

[12] ACRIS data are available through NYC Open Data. I use mortgage documents (type MTGE) recorded before June 2019.

[13] NYC assesses multifamily rental buildings (Class 2) at 45% of market value, so a mortgage-to-assessed ratio of 1.85 corresponds to an approximate market loan-to-value ratio of 83% at origination. See NYC Department of Finance, *Definitions of Property Assessment Terms*, `https://www.nyc.gov/site/finance/property/definitions-of-property-assessment-terms.page`.

[14] HPD Registration Contacts (`feu5-w2e2`) and Multiple Dwelling Registrations (`tesw-yqqr`) from NYC Open Data.

[15] The remaining 1,064 buildings (2.0% of the analysis sample) do not match to an HPD owner identifier. They remain in the main analysis but are not used in owner-portfolio classifications.

3.2 Sample and Summary Statistics

The analysis sample includes buildings with six or more residential units, constructed in 1990 or earlier, observed from July 2016 through December 2025. I restrict to older, larger buildings to focus on the multifamily rental stock where rent stabilization is prevalent and to ensure common support between stabilized and non-stabilized buildings. I exclude NYCHA public housing and buildings receiving Low-Income Housing Tax Credit (LIHTC), Department of Housing and Urban Development (HUD), or Mitchell-Lama subsidies,[16] which face federal or state affordability requirements distinct from rent stabilization. I also exclude condominiums, where individual unit owners, rather than a single landlord, make maintenance decisions; results including condos are nearly identical.

Table 1 summarizes the sample construction, showing how the final analysis sample is derived from the universe of NYC properties. The sample begins with all lots in the PLUTO database, then applies restrictions for multifamily buildings (6+ units), valid year built, construction in 1990 or earlier, and available community district information. NYCHA public housing and subsidized housing (LIHTC, HUD-assisted, Mitchell-Lama) are excluded because these buildings face distinct regulatory regimes.

Table 2 presents summary statistics separately for rent-stabilized and non-stabilized buildings. Stabilized buildings average 30 units compared with 32 for non-stabilized buildings, and the two groups are of similar vintage, with average construction years of 1920 and 1922. However, stabilized buildings have substantially higher baseline violation rates (0.16 Class C violations per month vs. 0.06) and complaint rates (0.55 vs. 0.17 per month). Figure 1 shows the geographic distribution: rent-stabilized buildings are concentrated in Manhattan, the Bronx, and parts of Brooklyn and Queens, with substantial overlap between stabilized and non-stabilized buildings within neighborhoods.

Both stabilized and non-stabilized buildings in the sample are predominantly prewar construction (92% and 88%, respectively), with similar distributions across walk-up and elevator building types. The non-stabilized buildings are largely prewar buildings that either decontrolled prior to HSTPA through vacancy decontrol or were never enrolled in stabilization. Only 4.6% of non-stabilized buildings were constructed between 1974 and 1990 (after stabilization began).

[16] I identify subsidized buildings using the HPD Affordable Building dataset from NYC Open Data.

4 Empirical Strategy

I estimate the effect of HSTPA by comparing changes in outcomes for rent-stabilized buildings with changes for non-stabilized buildings before and after the law took effect in June 2019. The identification challenge is that stabilized buildings were not comparable to the control group at baseline: they had nearly three times as many Class C violations per month (0.16 vs. 0.06) and more than three times as many complaints (0.55 vs. 0.17). These differences matter because buildings with worse baseline conditions may have followed different trends even without HSTPA.

To make the comparison credible, I condition on building fixed effects and a set of group-by-period fixed effects. Building fixed effects remove time-invariant differences across properties. Tract × period fixed effects compare stabilized and non-stabilized buildings facing the same local shocks, including changes in housing demand, gentrification, enforcement intensity, and neighborhood-level pandemic disruptions. Size × period, construction-decade × period, and building-class × period fixed effects allow physically different buildings to follow different time paths.[17]

Finally, baseline-outcome tercile × period fixed effects address the fact that stabilized and non-stabilized buildings differ substantially in baseline outcome levels (Table 2), and buildings at different quality levels likely evolve on different trajectories regardless of treatment. I compute baseline terciles from the three-year pre-HSTPA average separately for each outcome and interact them with period fixed effects, allowing each quality tier to follow its own path. Because terciles are computed from a multi-year average rather than a single lagged value, they primarily capture persistent building characteristics rather than transitory fluctuations. Thus, identification comes from comparing stabilized and non-stabilized buildings observed in the same period and local tract, while allowing buildings of different sizes, vintages, types, and baseline violation propensities to follow separate time paths. Appendix Table A2 shows why this adjustment matters: specifications without baseline-outcome × period controls exhibit clear pre-trends, while the preferred specification does not.

I estimate:

$$Y_{it} = \alpha_i + \sum_{k \neq -1} \beta_k \cdot \mathbf{1}[t = k] \cdot \text{Stabilized}_i + \gamma^{\text{tract}}_{c(i),t} + \gamma^{\text{size}}_{s(i),t} + \gamma^{\text{decade}}_{d(i),t} + \gamma^{\text{type}}_{b(i),t} + \gamma^{\text{baseline}}_{q(i),t} + \varepsilon_{it} \quad (1)$$

where Y_{it} is the outcome for building i in 6-month period t, α_i are building fixed effects, and the γ terms are the group-by-period fixed effects indexed by tract c, size bin s, construction

[17] Size bins are 6–10, 11–20, 21–50, 51–100, and 100+ units. Decade bins span pre-1920 through the 1990s. Building classes are walk-up apartments, elevator apartments, and mixed-use buildings.

decade d, building class b, and baseline-outcome tercile q. The baseline tercile $q(i)$ is defined separately for each outcome. The omitted period is January–June 2019, the final pre-HSTPA bin. The coefficients β_k measure the differential change in outcomes for stabilized buildings in period k relative to that omitted period. Standard errors are clustered at the building level to allow arbitrary serial correlation within the same building.

For average post-HSTPA effects, I also estimate the pooled difference-in-differences specification:

$$Y_{it} = \alpha_i + \beta \cdot \text{Stabilized}_i \times \text{Post}_t + \gamma^{\text{tract}}_{c(i),t} + \gamma^{\text{size}}_{s(i),t} + \gamma^{\text{decade}}_{d(i),t} + \gamma^{\text{type}}_{b(i),t} + \gamma^{\text{baseline}}_{q(i),t} + \varepsilon_{it} \quad (2)$$

where $\text{Post}_t = \mathbf{1}[t \geq 0]$ indicates periods after HSTPA. The coefficient β is the average post-HSTPA effect for stabilized buildings relative to non-stabilized buildings, conditional on the same fixed effects as in the event-study specification.

For count outcomes (violations, complaints, and permits), I express outcomes per 100 residential units and weight regressions by the number of residential units. The estimates therefore have a unit-weighted interpretation: they describe outcomes for the average housing unit rather than the average building. Appendix A.1 reports unweighted raw-count specifications, which produce similar results.

I aggregate observations into 6-month periods because monthly violations and complaints are sparse at the building level, and 6-month bins reduce noise while smoothing seasonal patterns such as winter heating complaints. The bins also align with HSTPA's timing: the law passed on June 14, 2019, so I assign June 2019 to the final pre-period bin (January–June 2019) and define July–December 2019 as the first post-period bin. For inspection-to-certification time, the 6-month outcome is the average of the building-month median intervals defined in Section 3.

As a robustness check, I also estimate a dose-response version of the specification that replaces the binary treatment indicator with the pre-HSTPA share of units that are rent-stabilized. This test asks whether effects are larger in buildings more exposed to the reform.

The key identifying assumption is conditional parallel trends: absent HSTPA, stabilized and non-stabilized buildings in the same fixed-effect cells would have experienced similar changes in outcomes. This assumption would fail if, within the same tract, size bin, construction cohort, building class, and baseline-outcome tercile, stabilized buildings were already on a different trajectory for reasons unrelated to HSTPA. The event-study coefficients provide a diagnostic for this threat. Pre-HSTPA coefficients close to zero indicate that treated and control buildings were moving similarly before the reform, though they cannot prove that post-HSTPA counterfactual trends would have remained parallel.

The design also assumes limited anticipation. HSTPA was debated before June 2019, so landlords may have adjusted behavior before passage. The pre-period event-study coefficients are close to zero across outcomes, which suggests little anticipatory response on the maintenance margin. If anything, the most plausible anticipatory response works against finding deterioration: the spike in MCI applications before HSTPA (Appendix Figure A2) suggests some landlords accelerated capital investment, which would leave stabilized buildings in better condition at the start of the post-period and attenuate the estimated effects. Treatment classification is also predetermined: I define stabilized buildings using the 2017 DHCR registry, two years before HSTPA's passage, so landlords could not alter treatment status in anticipation of or response to the law.

A final concern is spillovers across buildings owned by the same landlord: if landlords who own both stabilized and non-stabilized buildings reallocate resources between them after HSTPA, the control group would improve mechanically and the estimates would overstate deterioration in treated buildings. Section 5 tests this implication by comparing non-stabilized buildings owned by mixed-portfolio landlords with non-stabilized buildings owned by landlords without mixed portfolios.

5 Results

5.1 Main Results

Violations. Housing code violations provide a direct measure of building conditions, with Class C violations representing immediately hazardous conditions. Figure 2 presents event studies separately for Class B (hazardous) and Class C (immediately hazardous) violations. The pre-period coefficients are close to zero and statistically insignificant. The post-HSTPA increase is concentrated in the more serious violation classes, with the largest proportional increase for Class C violations.

Table 3 reports the pooled difference-in-differences estimates. Class C violations increased by 1.2 violations per 100 residential units per 6-month period, a 36% increase relative to the pre-HSTPA baseline of 3.2 Class C violations per 100 units for stabilized buildings. Class B violations increased by 0.73 per 100 units (8% of baseline). Figure 3 presents the corresponding event study for the time from inspection to certification. The pooled estimate increases by 35 days (34% of baseline), suggesting longer delays before cited problems are certified as corrected. This reduced-form effect may reflect slower landlord remediation, shifts in violation mix or severity, or changes in enforcement processes.

Complaints. Figure 4 presents the event study for tenant complaints. Complaints represent tenant-reported problems rather than inspector-identified violations, providing a complementary measure of building quality that does not depend on enforcement intensity.

Table 3 shows that HPD complaints increased by 1.4 per 100 units per 6-month period (13% of baseline). These findings address the concern that violation results might reflect changes in enforcement intensity: complaints are filed by tenants regardless of inspection activity.

HSTPA may have increased tenant willingness to report problems, which could account for part of the violation increase even without actual quality decline. Several patterns suggest this is not the primary driver. First, Class C violations are the most serious HPD hazard class, so tenants are unlikely to leave them unreported in any case. Second, the increases are largest for the most serious class and smallest for the least serious: Class C rose by 36%, Class B by 8%, and Class A by 4%. Increased reporting of marginal conditions would predict the opposite gradient. Third, the permit decline provides corroborating evidence that does not depend on tenant behavior: permits reflect landlord decisions filed with the Department of Buildings, and the 25% decline cannot be explained by changes in reporting. Fourth, complaints are the outcome that most directly captures tenant reporting, yet they rose by only 13%, far less than the 36% increase in Class C violations; if reporting were the primary channel, complaints should rise more than violations, not less. Together, these patterns point to real quality deterioration rather than solely increased reporting.

Permits. Figure 5 presents the event study for building permits. Alteration permits are required for significant construction work including structural modifications, major plumbing, and electrical work, and thus capture capital investment rather than routine maintenance. Permit activity declined by -0.22 permits per 100 units per 6-month period following HSTPA, a 25% decline relative to the baseline of 0.87 permits per 100 units.

The permit decline is smaller in percentage terms than the violation and complaint increases, but still economically meaningful.

Separately, MCI applications—which allow landlords to recover the cost of building-wide improvements through rent increases—collapsed by 81% immediately after HSTPA's passage. Because MCIs are available only for rent-stabilized buildings, there is no valid control group, so this is descriptive evidence only. Appendix B.1 presents the time series.

5.2 Robustness

Fixed effects specifications. Table A2 in Appendix A.2 estimates specifications with progressively richer fixed effects structures for Class C violations. All specifications produce positive point estimates, but simpler specifications exhibit substantial pre-trend concerns.

With only building fixed effects, pre-period coefficients are highly significant. Adding neighborhood and building characteristic controls progressively reduces pre-trend violations, but only the full specification with baseline-tercile × time fixed effects achieves clean pre-trends.

One concern with matching on baseline outcomes is mean reversion. The worry is that high-violation control buildings might naturally improve over time, making stabilized buildings look worse even if HSTPA had no effect. Appendix A.5 checks this by estimating event studies separately within baseline violation terciles. Within the middle and high baseline-risk groups, stabilized and non-stabilized buildings show little differential movement before HSTPA, but the gap widens afterward. This pattern provides evidence against differential mean reversion as the main explanation for the results.

Treatment intensity. If HSTPA caused the observed decline in maintenance quality, effects should scale with treatment intensity. Figure 6 tests this by comparing the binary treatment (any stabilization) with a continuous measure (share of units stabilized).

Both specifications show flat pre-trends followed by divergence after HSTPA. The continuous specification yields a larger point estimate, consistent with dose-response: fully stabilized buildings experience larger effects than partially stabilized buildings. Table A3 in Appendix A.3 confirms that this pattern holds across all outcomes. As a further check, Table A4 restricts the sample to stabilized buildings only, comparing buildings with more or fewer stabilized units. Effects remain significant across outcomes here as well.

Portfolio spillovers. A potential threat to identification arises if landlords who own both stabilized and non-stabilized buildings reallocate resources toward their non-stabilized properties after HSTPA, contaminating the control group—a violation of the stable unit treatment value assumption (SUTVA), under which one building's treatment status does not affect another building's outcomes.

To test for this, I link buildings to owners using HPD registration data (see Appendix C for details). I identify 1,880 "mixed-portfolio" landlords who own both stabilized and non-stabilized buildings, covering 16,572 analysis-sample buildings. I then compare non-stabilized buildings owned by mixed-portfolio landlords to those owned by all-market landlords. Restricting the sample to non-stabilized buildings only, I estimate:

$$y_{it} = \beta \cdot \text{Mixed}_i \times \text{Post}_t + \alpha_i + \gamma_{g(i),t} + \varepsilon_{it} \tag{3}$$

where $\text{Mixed}_i = 1$ if building i is owned by a landlord who also owns stabilized buildings. If reallocation were occurring, we would expect $\beta < 0$ (mixed-portfolio non-stabilized buildings improving after HSTPA).

Table 4 presents the results. If mixed-portfolio landlords reallocated resources toward their non-stabilized buildings after HSTPA, the coefficient would be negative: mixed-portfolio non-stabilized buildings would improve relative to all-market non-stabilized buildings. Instead, the estimate is positive and statistically insignificant, providing no evidence of this reallocation channel. As an additional check, I re-estimate the main specification on buildings with single-type owner portfolios, excluding mixed-portfolio and unmatched buildings; the effect remains positive and significant. These results reduce concern that the main estimates are driven by portfolio-level spillovers into the control group.

COVID-19. The post-period includes the COVID-19 pandemic, which disrupted HPD inspections, tenant complaint behavior, and construction activity. Several features of the research design address potential bias from COVID-related disruptions. First, the tract × period fixed effects absorb any citywide or neighborhood-level disruptions that affected all buildings equally. The identifying assumption requires only that COVID did not differentially affect stabilized versus non-stabilized buildings within the same neighborhood, a plausible assumption since inspection protocols applied uniformly. Second, as a robustness check, I exclude the COVID period entirely (January 2020 through December 2021). Results are similar across all outcomes (Appendix Table A5).

Sample composition. The main sample excludes condominiums, which have different maintenance incentive structures than rental buildings (individual unit owners make decisions rather than a single landlord). Including condos in the sample yields nearly identical results, confirming that the findings are robust to this sample restriction.

5.3 Heterogeneity

The aggregate effects may mask important variation across neighborhoods. To explore this, I estimate the main specification separately by borough, neighborhood income, and neighborhood non-white population share.

Table 5 presents pooled difference-in-differences results for Class C violations. Effects vary substantially across boroughs: the Bronx shows the largest increase (+2.51 violations per 100 units), more than twice Manhattan's effect (+1.02). Brooklyn (+1.07) and Queens (+0.95) fall in between.

Effects are larger in lower-income neighborhoods and in neighborhoods with higher non-white population shares. Splitting tracts into terciles by median household income, the effect in low-income neighborhoods (+1.72) is nearly three times the effect in high-income neighborhoods (+0.59). Splitting tracts by non-white population share, the effect in the top tercile (+2.28) is more than three times the effect in the bottom tercile (+0.64). Formal interaction tests reject equality between the lowest- and highest-income terciles and between the lowest and highest non-white-share terciles; they also show significantly larger effects in middle-income neighborhoods relative to high-income neighborhoods.

The concentration of effects in lower-income and predominantly non-white neighborhoods could reflect landlord financial constraints, differential maintenance behavior, differences in enforcement intensity, or other factors. Regardless of mechanism, the pattern suggests that the policy's unintended consequences fall disproportionately on disadvantaged communities.

I next ask whether these effects are larger for buildings more exposed to financial constraints. If reduced expected returns are the central mechanism, landlords with less financial slack may cut maintenance more sharply because they have less ability to absorb the decline in expected returns. I test this using pre-HSTPA mortgage leverage.

Table 6 tests whether effects vary with financial capacity using the mortgage-to-assessed ratio described in Section 3.[18] I estimate a triple-difference specification that interacts Stabilized $\times$ Post with indicators for leverage group. Buildings without a recorded pre-HSTPA mortgage form the reference group, while buildings with recorded mortgages are split into terciles of the mortgage-to-assessed ratio.

Buildings without a recorded pre-HSTPA mortgage show an increase of 0.58 Class C violations per 100 units. Effects are larger among buildings with recorded mortgages: low leverage (+1.09), middle leverage (+2.07), and high leverage (+1.62). The interaction terms comparing each leverage tercile to the no-recorded-mortgage group are all significant. The relationship is not strictly monotonic: middle-leverage buildings show the largest effects, possibly reflecting measurement error in the leverage proxy or the fact that the most highly leveraged buildings may already have been in financial distress before HSTPA. The pattern is consistent with leverage amplifying the maintenance response to HSTPA, while also showing that the effect is not limited to buildings with recorded pre-HSTPA mortgages.

6 Conclusion

This paper studies how strengthening rent control affects housing quality. I use New York's 2019 Housing Stability and Tenant Protection Act, which sharply reduced landlords' ability to raise rents and recover capital costs, and compare outcomes in rent-stabilized and non-stabilized buildings. I find that immediately hazardous violations per 100 units increased by 36%, complaints increased by 13%, and inspection-to-certification intervals increased

[18] This proxy is based on the most recent pre-HSTPA mortgage origination amount, not outstanding balances, so it captures relative leverage across buildings rather than true loan-to-value ratios. Buildings without a recorded pre-HSTPA mortgage form the reference group; they are not necessarily debt-free.

by 34%. Building permit activity also declined by 25%. These effects persist over the six-year study period. Effects concentrate in lower-income and predominantly non-white neighborhoods, where violations increased around three to four times more than in higher-income, predominantly white areas. These findings suggest that strengthening rent control reduced both routine maintenance and capital investment, with the largest effects in the most disadvantaged communities.

A full welfare analysis of HSTPA would need to weigh the costs documented here against the benefits to tenants of stronger rent protections and reduced displacement risk, along with the well-documented effects of rent control on supply and misallocation. The increase in hazardous violations represents a real cost to tenant well-being, but so do rent instability and the threat of eviction through decontrol. This paper does not resolve which effect dominates. What it does establish is that quality deterioration is a real and quantitatively meaningful cost of strengthening rent control, one that operates even when the supply margin is constrained by other factors. This cost should be weighed explicitly in policy design. Potential mitigations might include enhanced code enforcement, stronger incentives for building maintenance, reformed capital-cost recovery mechanisms, or targeted subsidies for buildings in disadvantaged neighborhoods where effects concentrate.

These results have immediate policy relevance. New York City is currently debating a rent freeze on its one million rent-stabilized apartments, and rent control policies are under active debate in many other cities. The findings here indicate that such policies, absent countervailing measures, could accelerate the quality decline documented in this paper. More broadly, the results suggest that municipalities considering stronger rent control should anticipate potential quality effects and consider complementary measures to address them, whether by maintaining monetary incentives for investment or by increasing enforcement. The concentration of effects in disadvantaged neighborhoods underscores that absent such measures, the costs may fall disproportionately on the communities rent control is intended to protect.

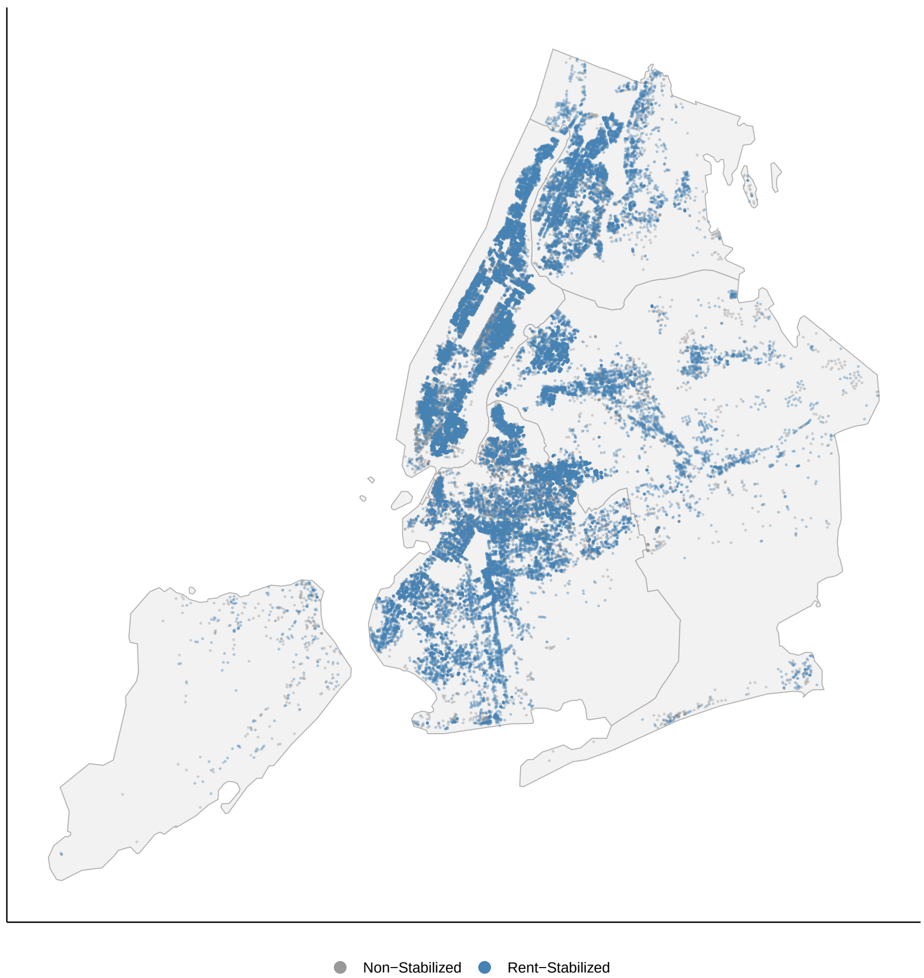
Figures and Tables

Table 1: Sample Construction

Sample Restriction	Buildings	Dropped
All NYC lots (PLUTO)	858,644	—
Multifamily ($\geq$6 units)	68,677	789,967
Valid year built	68,395	282
Built 1990 or earlier	57,189	11,206
Excl. NYCHA & subsidized	56,505	684
Excl. condos (R-class)	53,825	2,680

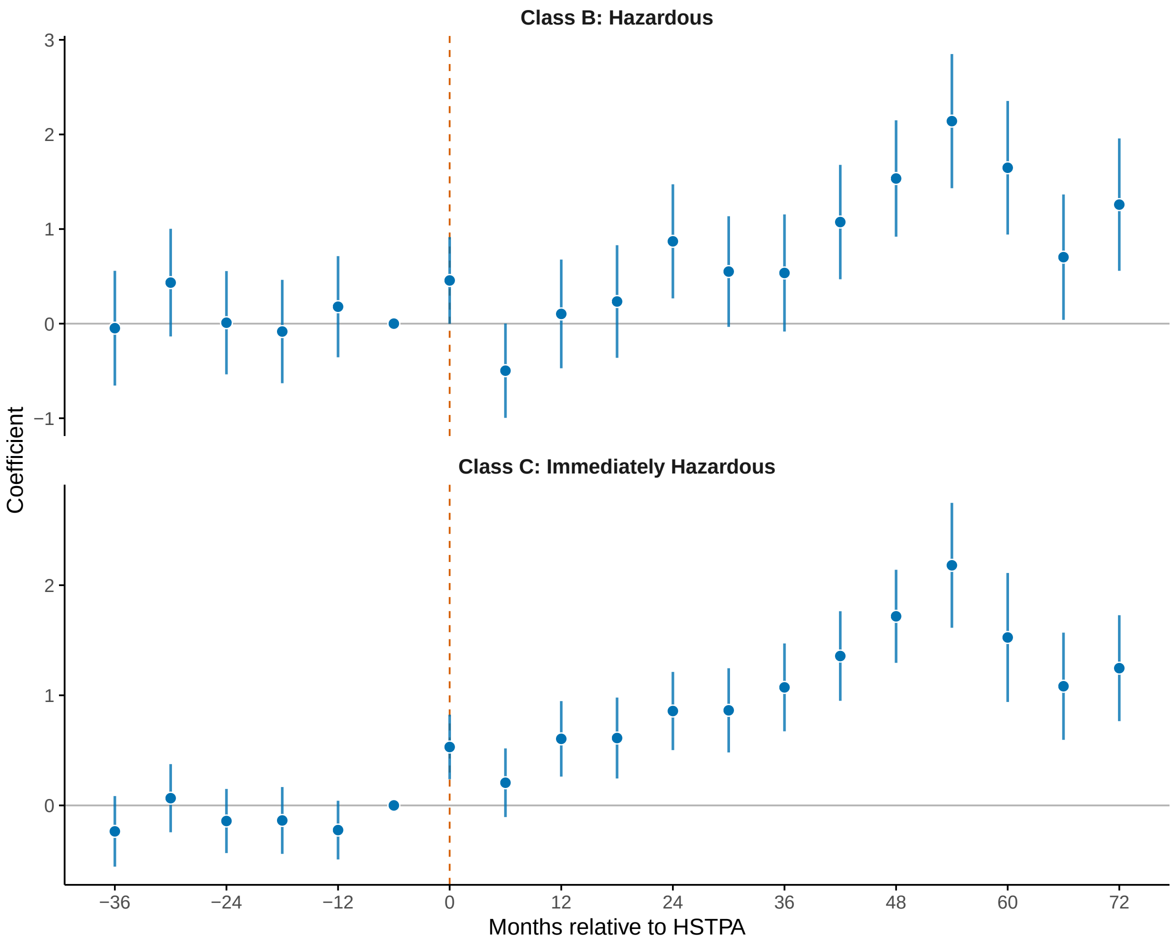
Note: Table shows the number of buildings remaining after each sample restriction. “Dropped” indicates the number of buildings excluded at each step. NYCHA refers to New York City Housing Authority public housing. Subsidized housing includes LIHTC, HUD-assisted, and Mitchell-Lama buildings.

Figure 1: Geographic Distribution of Sample Buildings by Rent Stabilization Status



Note: The map plots all analysis-sample buildings with valid PLUTO lot centroids. The analysis sample excludes NYCHA public housing, condominiums, and buildings receiving LIHTC, HUD, or Mitchell-Lama subsidies. Blue points indicate mapped rent-stabilized buildings; gray points indicate mapped non-stabilized buildings. The full analysis sample contains 38,992 rent-stabilized and 14,651 non-stabilized buildings. Rent-stabilization status is classified using the 2017 DHCR registry.

Figure 2: Event Study: Effect of HSTPA on Housing Code Violations



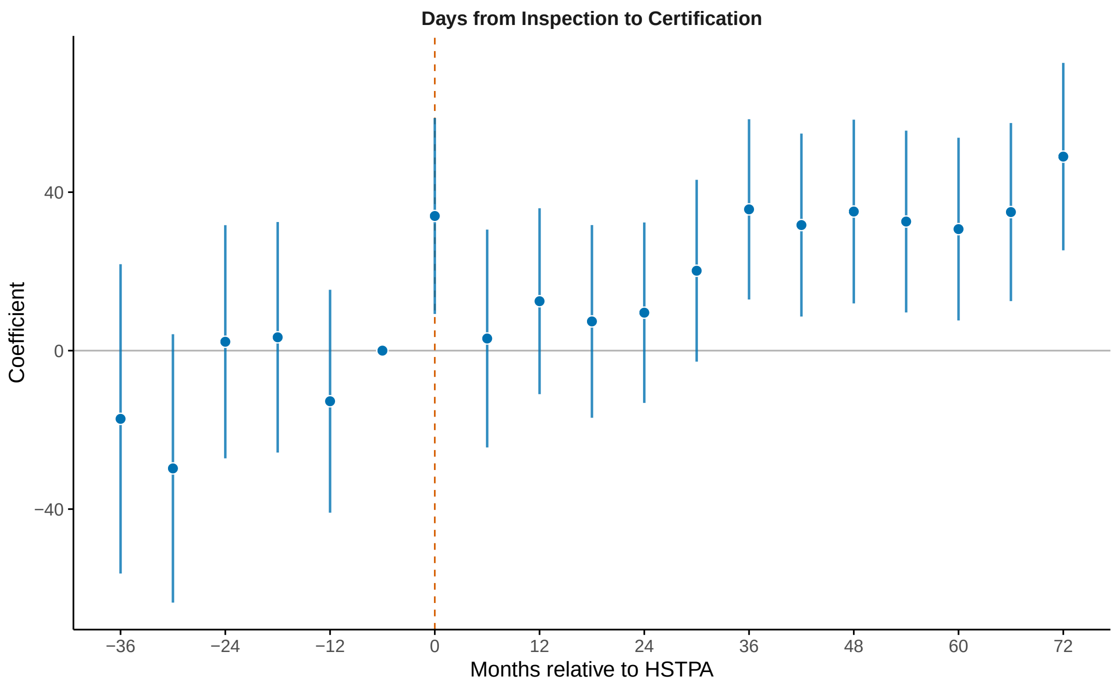
Note: Sample consists of 53,643 NYC buildings with six or more residential units built in 1990 or earlier, excluding NYCHA public housing, condominiums, and buildings receiving LIHTC, HUD, or Mitchell-Lama subsidies. Figure plots coefficients β_k on interactions between an indicator for rent-stabilized buildings and 6-month period indicators, relative to January–June 2019 (period $k = -1$). Outcomes are Class B (hazardous) and Class C (immediately hazardous) HPD violations per 100 residential units, weighted by building size. Specification includes building fixed effects plus tract × period, size-bin × period, decade × period, building-type × period, and baseline-tercile × period fixed effects. Vertical dashed line indicates HSTPA passage (June 2019). Error bars show 95 percent confidence intervals. Standard errors are clustered at the building level.

Table 2: Summary Statistics

	Stabilized		Non-Stabilized	
	Mean	SD	Mean	SD
Panel A: Building Characteristics				
Total Units	29.9	67.0	31.6	155.1
Year Built	1922	19	1920	25
Panel B: Pre-Period Outcomes (monthly)				
Class C Violations	0.161	0.414	0.056	0.260
Class B Violations	0.461	1.142	0.147	0.695
HPD Complaints	0.549	1.898	0.173	0.857
DOB Alterations	0.043	0.129	0.062	0.186
N Buildings	38,992		14,651	

Note: Sample consists of 53,643 NYC buildings with six or more residential units built in 1990 or earlier, excluding NYCHA public housing, condominiums, and buildings receiving LIHTC, HUD, or Mitchell-Lama subsidies. Rent stabilization status is determined from the 2017 DHCR registry. Violation and complaint data are from NYC HPD Open Data; permit data from NYC DOB Open Data. Outcome variables are measured at the building-month level during the pre-HSTPA period (July 2016 through June 2019). Class C violations are immediately hazardous conditions.

Figure 3: Event Study: Effect of HSTPA on Violation Certification Timing



Note: Sample consists of analysis-sample building-months with at least one HPD violation with a nonmissing, nonnegative interval from inspection to certification. Figure plots coefficients on interactions between an indicator for rent-stabilized buildings and 6-month period indicators, relative to January–June 2019. The outcome is the average within each 6-month period of the building-month median number of days from HPD inspection to certification of correction. Specification includes building fixed effects plus tract × period, size-bin × period, decade × period, building-type × period, and baseline-tercile × period fixed effects. Vertical dashed line indicates HSTPA passage (June 2019). Error bars show 95 percent confidence intervals. Standard errors are clustered at the building level.

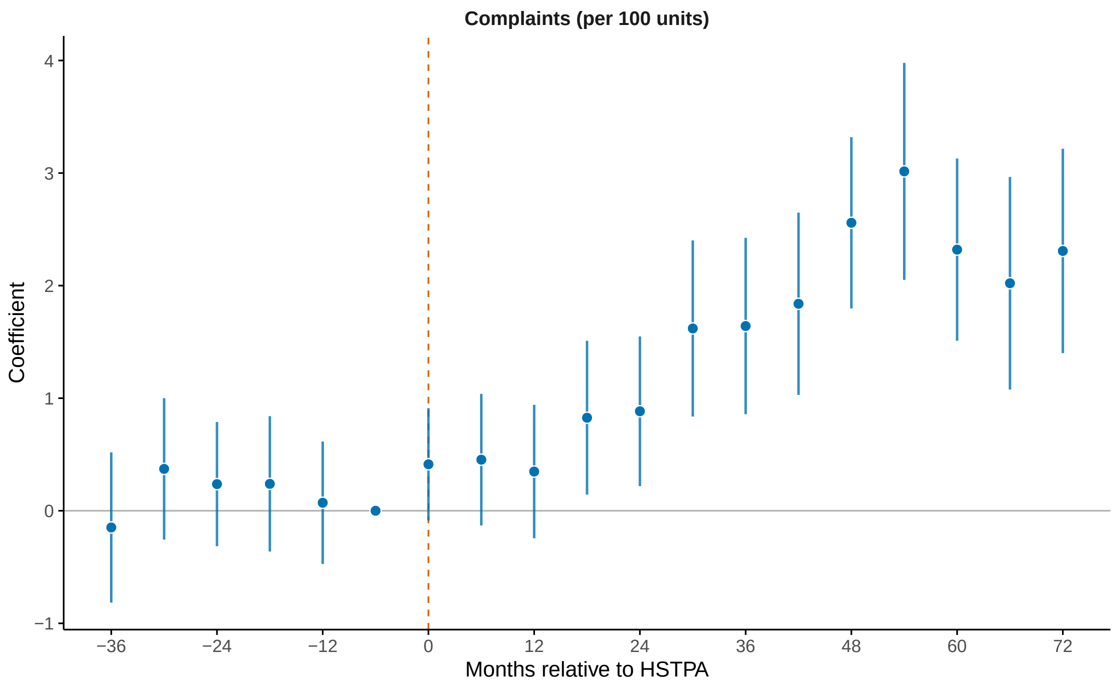
Table 3: Effect of HSTPA on Building Outcomes

	Class A	Class B	Class C	Complaints	Days to Cert.	Permits
Stabilized × Post	0.1375*	0.7343***	1.178***	1.429***	34.52***	-0.2174***
	(0.0728)	(0.1838)	(0.1124)	(0.2382)	(7.123)	(0.0293)
Baseline mean	3.84	9.27	3.24	11.04	100.18	0.87
Buildings	53,643	53,643	53,643	53,643	34,127	53,643
Observations	1,019,217	1,019,217	1,019,217	1,019,217	207,739	1,019,217

Note: This table tests whether HSTPA affected housing quality and investment outcomes in rent-stabilized buildings. Sample consists of 53,643 NYC buildings with six or more residential units built in 1990 or earlier, excluding NYCHA public housing, condominiums, and buildings receiving LIHTC, HUD, or Mitchell-Lama subsidies. Data span July 2016 through December 2025 in 6-month periods. Pooled difference-in-differences coefficients on Stabilized × Post. Count outcomes are per 100 residential units and unit-weighted; Days to Certification is measured in days and is not unit-weighted. Specification includes building fixed effects plus tract × period, size-bin × period, decade × period, building-type × period, and baseline-tercile × period fixed effects. "Baseline mean" is the pre-period (July 2016–June 2019) mean for stabilized buildings. Standard errors clustered at the building level in parentheses. Raw-count, building-weighted results for count outcomes appear in Table A1.

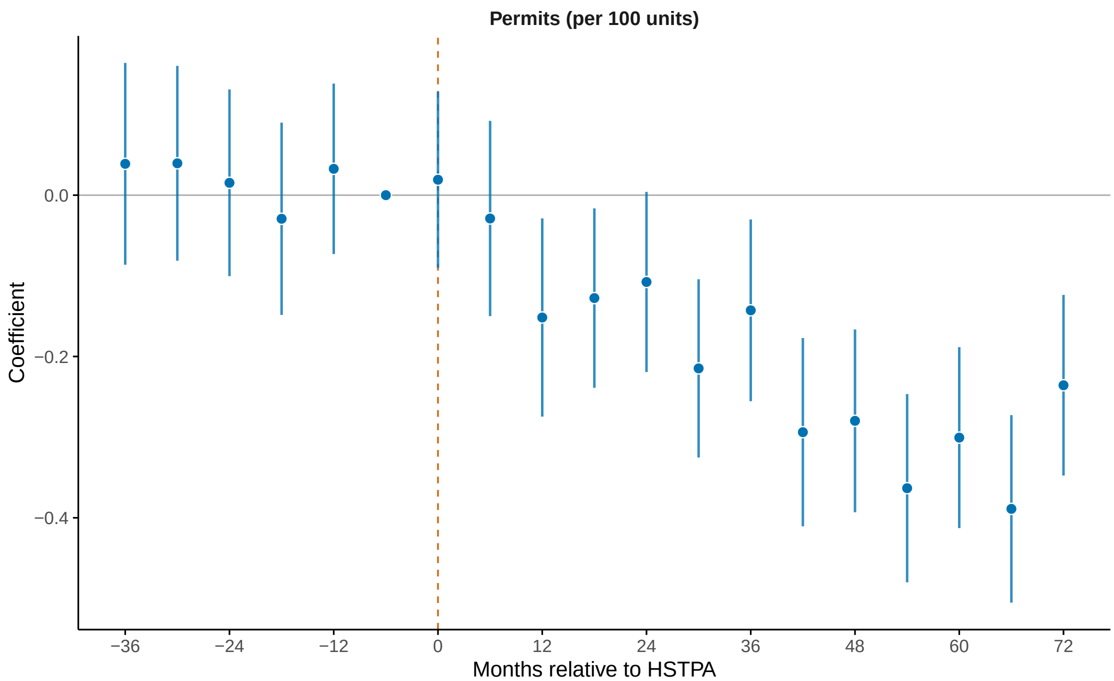
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Figure 4: Event Study: Effect of HSTPA on HPD Complaints



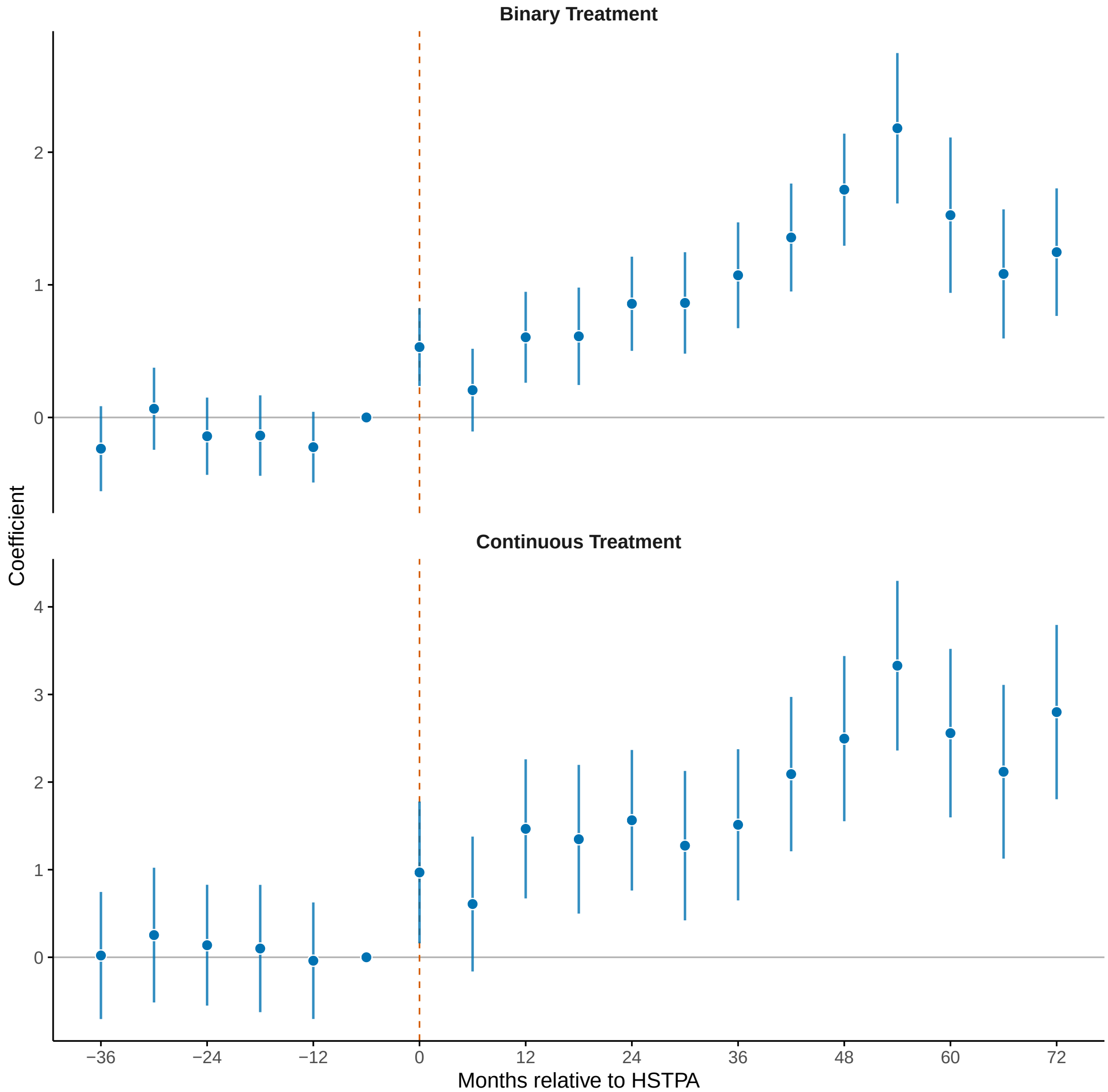
Note: Sample consists of 53,643 NYC buildings with six or more residential units built in 1990 or earlier, excluding NYCHA public housing, condominiums, and buildings receiving LIHTC, HUD, or Mitchell-Lama subsidies. Figure plots coefficients on interactions between an indicator for rent-stabilized buildings and 6-month period indicators, relative to January–June 2019. Complaints are normalized to per-100-units and weighted by building size. Specification includes building fixed effects plus tract × period, size-bin × period, decade × period, building-type × period, and baseline-tercile × period fixed effects. Vertical dashed line indicates HSTPA passage (June 2019). Error bars show 95 percent confidence intervals. Standard errors are clustered at the building level.

Figure 5: Event Study: Effect of HSTPA on Building Permit Activity



Note: Sample consists of 53,643 NYC buildings with six or more residential units built in 1990 or earlier, excluding NYCHA public housing, condominiums, and buildings receiving LIHTC, HUD, or Mitchell-Lama subsidies. Figure plots coefficients on interactions between an indicator for rent-stabilized buildings and 6-month period indicators, relative to January–June 2019. Outcome is DOB alteration permits (types A1, A2, A3) per 100 residential units, weighted by building size. Specification includes building fixed effects plus tract × period, size-bin × period, decade × period, building-type × period, and baseline-tercile × period fixed effects. Vertical dashed line indicates HSTPA passage (June 2019). Error bars show 95 percent confidence intervals. Standard errors are clustered at the building level.

Figure 6: Binary vs. Continuous Treatment



Note: This figure compares binary and continuous treatment specifications for Class C violations per 100 units. Sample consists of 53,643 NYC buildings with six or more residential units built in 1990 or earlier, excluding condominiums. Coefficients are from 6-month period indicators interacted with treatment, relative to January–June 2019 (period $k = -1$). Top panel uses a binary indicator for rent-stabilized buildings from the 2017 DHCR registry. Bottom panel uses continuous treatment intensity (share of stabilized units, 0–1) from 2018 property tax bills. Both specifications include building fixed effects plus tract $\times$ period, size-bin $\times$ period, decade $\times$ period, building-type $\times$ period, and baseline-tercile $\times$ period fixed effects. Error bars show 95 percent confidence intervals with standard errors clustered at the building level.

Table 4: Portfolio Spillovers Test (SUTVA)

	Class C Violations
Mixed $\times$ Post	0.432
	(0.299)
Baseline mean (mixed)	3.08
Buildings (mixed)	4,788
Buildings (all-market)	8,926
Observations	260,566

Note: This table tests whether landlords reallocate resources from stabilized to non-stabilized buildings after HSTPA (a potential SUTVA violation). Sample consists of non-stabilized buildings with identified ownership, drawn from the analysis sample excluding NYCHA public housing, condominiums, and buildings receiving LIHTC, HUD, or Mitchell-Lama subsidies (see Appendix C). "Mixed" buildings are owned by landlords who own both stabilized and non-stabilized properties; "all-market" buildings are owned by landlords with only non-stabilized properties. The coefficient shows the differential change in Class C violations per 100 units for mixed-portfolio buildings relative to all-market buildings after HSTPA. A negative coefficient would be consistent with reallocation toward non-stabilized properties. Specification includes building fixed effects plus tract $\times$ period, size-bin $\times$ period, decade $\times$ period, building-type $\times$ period, and baseline-tercile $\times$ period fixed effects. Standard errors clustered at the building level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 5: Heterogeneity in Class C Violation Effects

Dimension	Group	Buildings	Baseline	Effect
Borough	Manhattan	18,289	2.39	1.023***
				(0.096)
	Bronx	7,178	5.92	2.510***
				(0.594)
	Brooklyn	19,630	4.22	1.068***
				(0.242)
	Queens	8,146	0.95	0.952***
				(0.228)
Nbhd Income	Low	19,279	5.46	1.719***
				(0.301)
	Middle	15,397	2.88	1.354***
				(0.222)
	High	18,890	1.01	0.591***
				(0.067)
Non-white	Low	17,899	0.97	0.639***
				(0.075)
	Middle	17,922	2.52	0.790***
				(0.155)
	High	17,810	5.82	2.284***
				(0.345)

Note: This table tests whether HSTPA's effects vary across borough and neighborhood characteristics. Sample consists of 53,643 NYC buildings with six or more residential units built in 1990 or earlier, excluding NYCHA public housing, condominiums, and buildings receiving LIHTC, HUD, or Mitchell-Lama subsidies. Each row reports a separate regression on the indicated subsample. Data span July 2016 through December 2025 in 6-month periods. "Effect" is the coefficient on Stabilized × Post; "N" is the number of buildings in that subsample. Neighborhood income and non-white population share are tract-level measures from the 2019 ACS 5-year estimates (2015–2019); terciles are defined within the estimation sample. All specifications include building fixed effects plus tract × period, size-bin × period, decade × period, building-type × period, and baseline-tercile × period fixed effects. Outcome: Class C violations per 100 units, unit-weighted. Standard errors clustered at building level.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 6: Leverage Triple Difference

Leverage Group	Total Effect	(SE)	Interaction	(SE)
No mortgage	0.578^{***}	(0.134)	—	
Low	1.086^{***}	(0.133)	0.508^{***}	(0.135)
Middle	2.071^{***}	(0.171)	1.493^{***}	(0.171)
High	1.616^{***}	(0.241)	1.038^{***}	(0.242)

Note: This table tests whether financial leverage amplifies HSTPA's effect on housing quality. Sample consists of NYC buildings with six or more residential units built in 1990 or earlier, excluding NYCHA public housing, condominiums, and buildings receiving LIHTC, HUD, or Mitchell-Lama subsidies. Data span July 2016 through December 2025 in 6-month periods. Outcome: Class C violations per 100 units, unit-weighted. "Total Effect" is the HSTPA effect (Stabilized × Post) for each leverage group. "Interaction" tests whether that group shows additional deterioration relative to buildings without a recorded pre-HSTPA mortgage. Leverage is the mortgage-to-assessed ratio using the most recent pre-HSTPA mortgage from ACRIS and 2018 assessed values from PLUTO; terciles defined among mortgaged buildings. All specifications include building fixed effects plus tract × period, size-bin × period, decade × period, building-type × period, and baseline-tercile × period fixed effects. Standard errors clustered at the building level.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

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A Additional Robustness Checks

A.1 Building-Weighted Count Results

The main specification reports violations, complaints, and permits as rates per 100 residential units and weights buildings by the number of units. That specification answers a unit-level question: how much did outcomes change for the average housing unit in a stabilized building?

This appendix instead keeps the outcomes as raw six-month counts and gives each building equal weight. The coefficient is therefore the additional number of violations, complaints, or alteration permits in the average stabilized building per six-month period after HSTPA, relative to comparable non-stabilized buildings. This is a robustness check for whether the main results are driven by large stabilized buildings receiving more weight.

Table A1 reports these building-weighted estimates. The signs and significance match the main results: violations and complaints increase while alteration permits fall, and the percentage effects relative to baseline are comparable. The estimates are smaller in levels only because they are measured per building rather than per 100 residential units, leaving the substantive conclusion unchanged.

Table A1: Effect of HSTPA on Building-Weighted Count Outcomes

	Class A	Class B	Class C	Complaints	Alterations
Stabilized × Post	0.0590***	0.1834***	0.2709***	0.2900***	-0.0535***
	(0.0152)	(0.0417)	(0.0273)	(0.0539)	(0.0064)
Baseline mean	1.15	2.77	0.97	3.30	0.26
Buildings	53,643	53,643	53,643	53,643	53,643
Observations	1,019,217	1,019,217	1,019,217	1,019,217	1,019,217

Note: This table reports pooled difference-in-differences estimates using raw six-month counts per building. The coefficient is the change in the number of violations, complaints, or alteration permits for the average stabilized building after HSTPA, relative to comparable non-stabilized buildings. Sample consists of 53,643 NYC buildings with six or more residential units built in 1990 or earlier, excluding NYCHA, subsidized housing, and condominiums. Data span July 2016 through December 2025 in 6-month periods. Each building is weighted equally, in contrast to the main specification where count outcomes are normalized per 100 units and weighted by building size. Specification includes building fixed effects plus tract × period, size-bin × period, decade × period, building-type × period, and baseline-tercile × period fixed effects. Standard errors clustered at the building level in parentheses.

$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

A.2 Fixed Effects Specification

Table A2 estimates specifications with progressively richer fixed effects structures for Class C violations, demonstrating that baseline-tercile matching is essential for achieving parallel pre-trends.

Table A2: Robustness to Alternative Fixed Effects Specifications

	Bldg	+Tract	+Size	+Decade	+Class	+Baseline
Stabilized × Post	3.442***	1.477***	1.377***	1.309***	1.265***	1.178***
	(0.0665)	(0.1093)	(0.1088)	(0.1095)	(0.1110)	(0.1124)
Building	Yes	Yes	Yes	Yes	Yes	Yes
Tract × Time		Yes	Yes	Yes	Yes	Yes
Size Bin × Time			Yes	Yes	Yes	Yes
Decade Built × Time				Yes	Yes	Yes
Building Class × Time					Yes	Yes
Baseline Outcome × Time						Yes
Buildings	53,643	53,643	53,643	53,643	53,643	53,643
Pre-trend max$\|t\|$	24.97	6.180	5.600	5.580	5	1.650
Observations	1,019,217	1,019,217	1,019,217	1,019,217	1,019,217	1,019,217

Note: Sample consists of 53,643 NYC buildings with six or more residential units built in 1990 or earlier, excluding NYCHA public housing, condominiums, and buildings receiving LIHTC, HUD, or Mitchell-Lama subsidies. Data span July 2016 through December 2025 in 6-month periods. Table reports pooled post-period coefficients from difference-in-differences regressions of Class C violations per 100 units (unit-weighted) on Stabilized × Post, where Post equals one for periods after June 2019. Columns differ in fixed effects structure, building progressively from building fixed effects only (column 1) to the full matching specification (column 6). "Pre-trend max$|t|$" is the maximum absolute t-statistic among pre-period event study coefficients; values above 2 indicate pre-trend concerns. Standard errors (in parentheses) are clustered at the building level.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

A.3 Continuous Treatment (Dose-Response)

This appendix presents the full dose-response results summarized in Section 5. Table A3 replaces the binary treatment indicator with the continuous share of rent-stabilized units (0–1, from 2018 property tax bills). The coefficient represents the effect of moving from 0% to 100% stabilized.

Table A4 restricts the sample to stabilized buildings only, using within-group variation in stabilization intensity as treatment.

Table A3: Effect of Stabilization Share on Building Outcomes

	Class C	Class B	Complaints	Permits
Stab. Share × Post	1.778***	2.120***	2.386***	-0.124***
	(0.230)	(0.320)	(0.371)	(0.040)
Baseline mean	4.70	12.26	13.01	1.14
Buildings	53,643	53,643	53,643	53,643

Note: This table tests whether effects scale with treatment intensity by using the continuous share of rent-stabilized units (0–1, from 2018 property tax bills) rather than the binary indicator. The coefficient represents the effect of moving from 0% to 100% stabilized. Sample consists of 53,643 NYC buildings with six or more residential units built in 1990 or earlier, excluding NYCHA, subsidized housing, and condominiums. Data span July 2016 through December 2025 in 6-month periods. All specifications include building fixed effects plus tract × period, size-bin × period, decade × period, building-type × period, and baseline-tercile × period fixed effects. Reported outcomes are per 100 residential units and weighted by building size (total units). Standard errors clustered by building.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

A.4 COVID-19 Period

Table A5 compares main results with and without the COVID-19 period (January 2020 through December 2021). Results are similar across all outcomes, indicating that the main findings are not driven by pandemic-related disruptions.

A.5 Mean Reversion Check

A potential concern with baseline-tercile matching is regression to the mean. The worry is simple: some non-stabilized buildings may appear in high-violation groups because they had a temporary bad spell before HSTPA. If those control buildings naturally improve over time, while stabilized buildings in the same group do not, the event study could show a positive post-HSTPA gap even without a causal effect of the law.

To test for this, I estimate event studies separately within each baseline violation tercile. This asks whether stabilized and non-stabilized buildings that started with similar violation histories were already drifting apart before HSTPA. If mean reversion were driving the results, that drift should be visible before the law changed, especially in the high-baseline group.

Figure A1 presents the results. The middle and high baseline terciles show little evidence of differential pre-HSTPA drift, while treatment-control gaps widen after HSTPA. The lowest tercile is less informative as a pre-trend test because Class C violations are sparse and

Table A4: Effect of HSTPA Within Stabilized Buildings

	Class C	Class B	Complaints	Permits
Stab. Share × Post	0.876**	2.255***	1.814***	-0.069
	(0.350)	(0.490)	(0.587)	(0.051)
Baseline mean	4.70	12.26	13.01	1.14
Buildings	38,992	38,992	38,992	38,992

Note: This table restricts the analysis sample to stabilized buildings only, using variation in the share of rent-stabilized units (from 2018 property tax bills) as a continuous treatment. The underlying analysis sample excludes NYCHA public housing, subsidized housing, and condominiums. The coefficient represents the effect of moving from 0% to 100% stabilized among buildings already classified as stabilized. This design eliminates the non-stabilized control group entirely, addressing concerns about control group contamination from previously decontrolled buildings. Specification otherwise identical to Table A3. Reported outcomes are per 100 residential units and weighted by building size (total units). Standard errors clustered by building.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

many buildings have zero baseline violations. Overall, the figure suggests that the main estimates are not driven by high-violation control buildings mechanically reverting toward lower violation rates.

B Supplementary Evidence

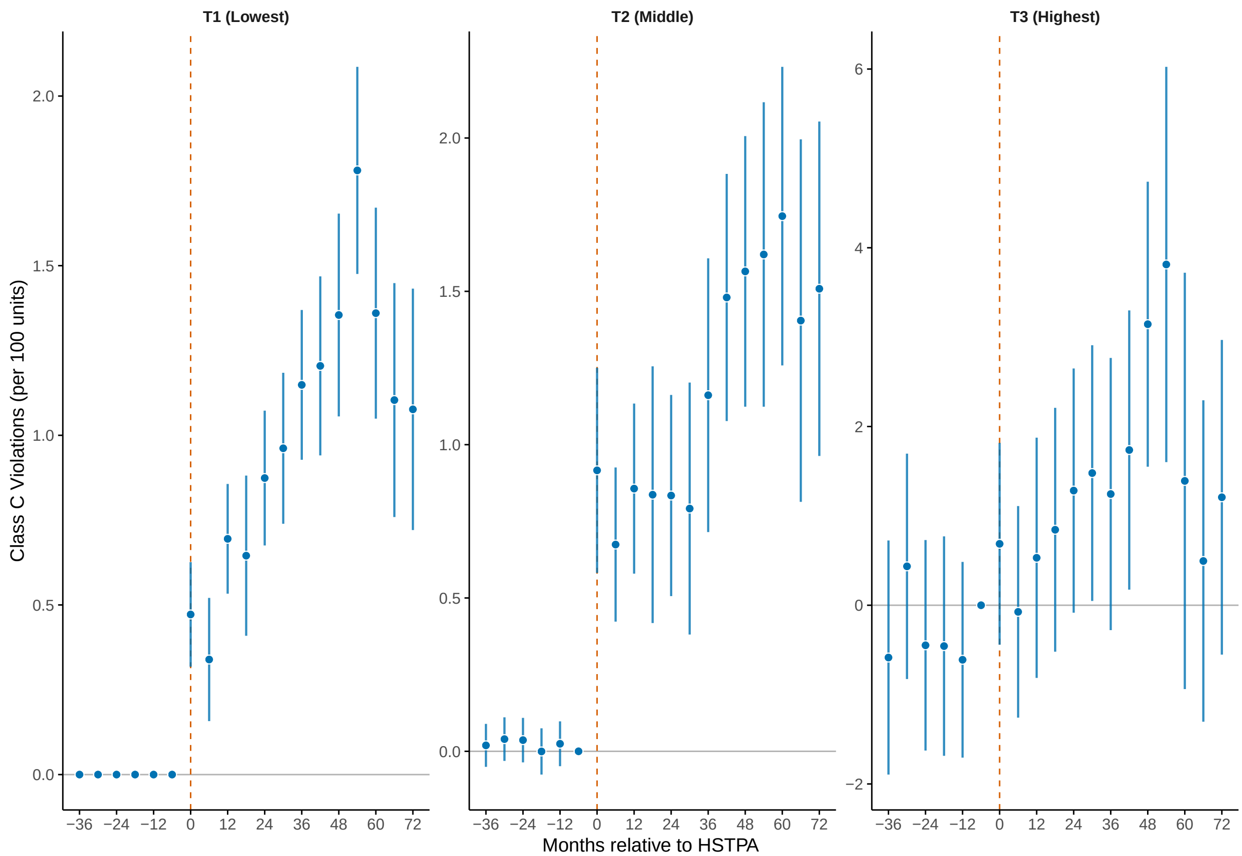
B.1 MCI Applications

Figure A2 shows the time series of Major Capital Improvement (MCI) applications. MCI filings collapsed immediately upon HSTPA's passage, falling from an average of 118 applications per month before HSTPA to 23 per month after, an 81% decline.

The June 2019 spike (313 applications, versus 118 monthly average) suggests anticipation: landlords who were planning MCI projects rushed to file before HSTPA reduced the return on such investments. The immediate collapse in July 2019 demonstrates that landlords respond quickly to changes in MCI economics.

The MCI collapse (81%) is not directly comparable to the permit decline (25%): the former is an unconditional pre/post change for stabilized buildings, while the latter is a difference-in-differences estimate relative to control buildings. The larger MCI decline likely reflects that MCIs are the rent-recovery mechanism (landlords file MCIs specifically to

Figure A1: Event Studies by Baseline Violation Tercile



Note: This figure tests whether differential mean reversion drives the results by estimating event studies separately within each baseline violation tercile. Sample consists of 53,643 NYC buildings with six or more residential units built in 1990 or earlier, excluding NYCHA, subsidized housing, and condominiums. Coefficients are from 6-month period indicators, relative to January–June 2019 (period $k = -1$), for Class C violations per 100 units. Terciles are formed by ranking buildings by baseline Class C violations, using BBL as a deterministic tie-breaker because baseline violations are sparse. Each specification includes building fixed effects plus tract $\times$ period, size-bin $\times$ period, decade $\times$ period, and building-type $\times$ period fixed effects (baseline-tercile $\times$ period is omitted since estimation is within-tercile). Unit-weighted. Vertical dashed line indicates HSTPA passage. Error bars show 95% confidence intervals with standard errors clustered at the building level.

Table A5: COVID-19 Period Robustness

Outcome	Full Sample	Excl. 2020–21
Class C Violations	1.178***	1.398***
	(0.112)	(0.133)
All Complaints	1.429***	1.842***
	(0.238)	(0.271)
Days to Certification	34.523***	40.801***
	(7.123)	(7.535)
Permits	-0.217***	-0.261***
	(0.029)	(0.031)
Buildings	53,643	53,643

Note: Comparison of main results including vs. excluding the COVID-19 pandemic period (January 2020–December 2021). Sample consists of 53,643 NYC buildings with six or more residential units built in 1990 or earlier, excluding NYCHA, subsidized housing, and condominiums. All specifications include building fixed effects plus tract $\times$ period, size-bin $\times$ period, decade $\times$ period, building-type $\times$ period, and baseline-tercile $\times$ period fixed effects. All count outcomes are per 100 residential units and unit-weighted; Days to Certification is measured in days and is not unit-weighted. Standard errors clustered at building level.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

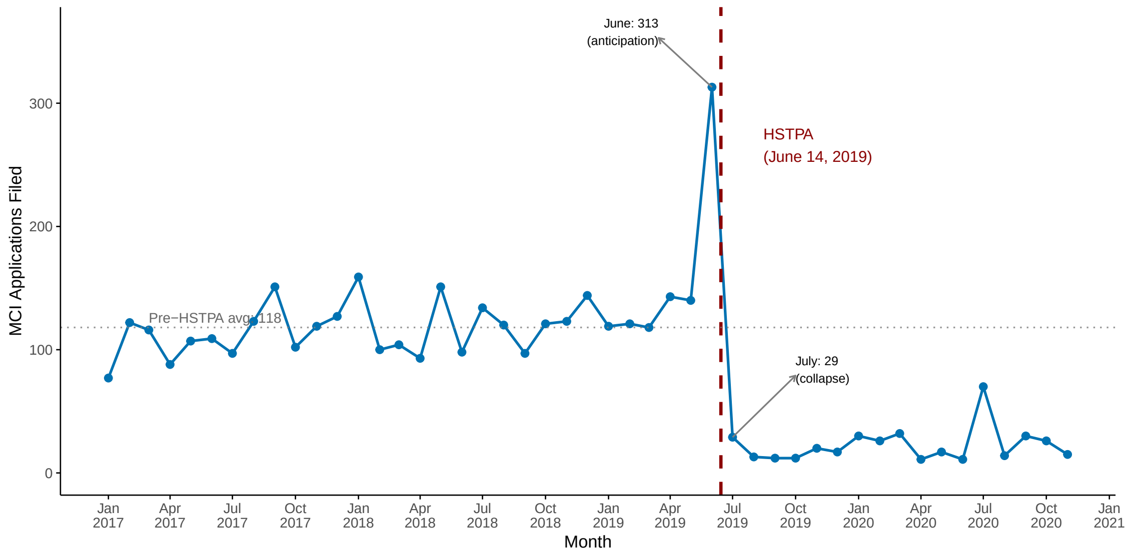
recoup capital costs through rent increases), while permits are just regulatory approval for construction work.

B.2 Violation Severity Gradient

If the violation increases reflected enhanced tenant reporting rather than actual quality decline, one would expect the largest increases for minor violations (which tenants might previously have tolerated) and smaller increases for serious violations (which tenants would report in any case). The data show the opposite pattern.

Table A6 presents effects by violation severity class. HPD classifies violations into three standard hazard classes: Class A (non-hazardous), Class B (hazardous), and Class C (immediately hazardous). Class C violations increased by 36% relative to baseline, Class B increased by 8%, and Class A increased by 4%. This gradient is consistent with real quality deterioration concentrated in the most serious deficiencies, and inconsistent with a pure

Figure A2: MCI Applications Over Time



Note: Figure shows monthly Major Capital Improvement (MCI) application filings to the Division of Housing and Community Renewal (DHCR) from January 2017 through November 2020. MCIs are building-wide improvements (boilers, roofs, elevators) for which landlords of rent-stabilized buildings can seek rent increases. Vertical dashed line indicates HSTPA passage (June 14, 2019). Pre-HSTPA monthly average: 118 filings. Post-HSTPA monthly average: 23 filings (81% decline). The June 2019 spike (313 filings) reflects anticipation as landlords rushed to file before the law took effect. Data are HCR/DHCR application records obtained by Winnie Shen through a FOIL request and distributed through NYCDB.

reporting-bias explanation.

C Landlord Matching and Portfolio Classification

To test for portfolio spillovers, I link buildings to individual owners using HPD Multiple Dwelling Registration data. NYC law requires owners of buildings with three or more residential units to register annually with HPD, providing contact information for the building owner. I use this data to identify which buildings share common ownership.

C.1 Data Source

The HPD Registration Contacts dataset lists contact persons for each registered building, including their role (e.g., HeadOfficer, IndividualOwner, Officer, Agent). The key challenge is that many buildings are owned through LLCs, which obscure the identity of the actual owner. However, HPD requires registration of the “head officer” or “individual owner” behind the LLC, providing person-level identifiers that can link buildings across different LLC structures.

I obtain the data from the NYC Open Data portal and link it to buildings using the

Table A6: HSTPA Effects by Violation Severity Class

Class	Severity	Deadline	Effect	% of Baseline
C	Immediately hazardous	Varies	1.2***	36%
B	Hazardous	30 days	0.73***	8%
A	Non-hazardous	90 days	0.14*	4%

Note: This table compares HSTPA effects across violation severity classes. Sample consists of 53,643 NYC buildings with six or more residential units built in 1990 or earlier, excluding NYCHA, subsidized housing, and condominiums. Effects are from pooled difference-in-differences specifications with building fixed effects plus tract × period, size-bin × period, decade × period, building-type × period, and baseline-tercile × period fixed effects. Outcomes are violations per 100 units, unit-weighted. Significance stars are based on standard errors clustered at the building level.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

HPD Multiple Dwelling Registrations dataset, which provides the BBL identifier for each registration.

C.2 Owner Name Standardization

For each building registration, I extract the owner name using the following priority order:

1. IndividualOwner (highest priority)
2. JointOwner
3. HeadOfficer
4. Officer (lowest priority)

I construct a standardized owner identifier by concatenating first name, middle initial, and last name, converting to uppercase and removing extra whitespace. This approach links buildings owned by the same person even when held through different LLCs.

C.3 Name Deduplication

Matching on full names (first, middle, and last) avoids false positives but may fail to link the same person when names are recorded inconsistently across registrations—for example, "Steven Croman" on one building and "Steven R Croman" on another. A naive solution would merge all names sharing the same first and last name, treating middle names as optional.

However, this creates substantial false positives for names where the middle component is a given name rather than an initial. For instance, "Mei Xian Chen" and "Mei Feng Chen" are distinct individuals with different given names, not spelling variants of the same person. Merging on first and last name alone would incorrectly combine thousands of distinct owners, particularly in Chinese, South Asian, and Hispanic communities where such naming patterns are common.

To balance accuracy against false positives, I implement a conservative deduplication procedure that requires direct evidence of common ownership. I identify name variants with identical first and last names that appear on the same building. When the same building lists both "Steven Croman" and "Steven R Croman" as contacts, this provides strong evidence they are the same person, and I merge all instances of both variants across the dataset. Approximately 4,400 buildings have multiple registration records, providing the overlap needed to identify such variant pairs. I also apply fuzzy string matching (requiring 85% character-level similarity) to names appearing on the same building, capturing minor spelling variations like "Hirschfield" versus "Hirscfield." This approach is conservative: it may fail to link name variants when no building happens to list both spellings.

C.4 Portfolio Classification

I merge the owner identifiers to the analysis panel and classify each owner's portfolio:

- **All-stabilized:** All of the owner's linked analysis-sample buildings are rent-stabilized
- **All-market:** None of the owner's linked analysis-sample buildings are rent-stabilized
- **Mixed:** The owner's linked analysis-sample buildings include both stabilized and non-stabilized buildings

Table A7 summarizes the portfolio classification. Of the 24,933 unique owners identified, 1,880 (7.5%) own mixed portfolios covering 16,572 analysis-sample buildings. These mixed-portfolio landlords are the focus of the SUTVA test, as they are the only owners who could potentially reallocate resources between stabilized and non-stabilized properties.

Table A7: Landlord Portfolio Classification

Portfolio Type	Owners	Buildings
All-stabilized	15,228	27,081
All-market	7,825	8,926
Mixed	1,880	16,572
Total with owner info	24,933	52,579
Unknown (no owner match)	—	1,064

Note: This table summarizes landlord portfolio types for buildings in the analysis sample, which excludes NYCHA public housing, subsidized housing, and condominiums. Owner identities are derived from HPD registration contacts using the name standardization procedure described in this appendix. A portfolio is "all-stabilized" if all of an owner's buildings are rent-stabilized, "all-market" if none are stabilized, and "mixed" if the owner holds both types. The Buildings column counts unique analysis-sample buildings assigned to each portfolio type, not owner-building links. Buildings that could not be linked to an owner are excluded from the SUTVA analysis (Table 4).